

Roger Andrés Álvarez Díaz

Front-End Developer (React)

Colombia | rogeralvarez22082007@gmail.com | +57 313 730 4143 | [GitHub](#) | [LinkedIn](#)

PROFESSIONAL PROFILE

Front-end focused developer with multiple shipped React applications built on top of real backend data layers (Flask APIs, OCR/CV pipelines, ML models). Comfortable translating data models and business logic into clean, functional UI, and integrating frontend components directly with REST APIs. Currently deepening TypeScript and exploring low-code/structured-data platforms; looking for a Front-End/Full-Stack role to grow into a polished, production-grade UI engineering practice.

TECHNICAL SKILLS

Front-End: React · JavaScript (ES6+) · HTML5 · CSS3 · Tailwind CSS · Vite

Back-End / Data: Python (Flask) · Node.js · REST APIs · SQL / MySQL / PostgreSQL

Tools & Practices: Git / GitHub · Postman · Scrum · Currently learning: TypeScript

FEATURED PROJECTS

Control RASS — Residential Access Control System — *React, Python, Flask, OpenCV*

- Problem: simulate automated vehicle access control with real-time license plate detection for an academic project.
- Solution: built the full React frontend consuming a Flask + OpenCV backend API, rendering live detection data and access events in a responsive UI.
- Impact: shipped a working end-to-end app (UI + API + CV pipeline) with documented architecture, published on GitHub.

Aquatic-Plants-Classfier-AI — *React, Vite, Flask, TensorFlow/EfficientNetB0*

- Problem: needed an accessible interface for non-technical users to classify aquatic plant species via an ML model.
- Solution: built a React/Vite frontend on top of a Flask API serving an EfficientNetB0 classifier, including Grad-CAM explainability visualizations rendered directly in the UI.
- Impact: delivered a fully documented, public-facing app connecting a trained ML model to a polished web interface.

smart-residential-security — *React/JavaScript, OCR*

- Problem: automate vehicle entry management for residential communities using license-plate recognition.
- Solution: built the frontend interface for intelligent entry management on top of an OCR-based detection pipeline.
- Impact: working access-control interface with real-time entry/exit handling logic.

graph-theory-robot-navigation (NAVBOT) — *JavaScript Canvas, Flask, NetworkX*

- Problem: visualize and compare pathfinding algorithms (Dijkstra, Bellman-Ford, Kruskal, TSP) interactively.
- Solution: built a canvas-based interactive frontend consuming a Flask + NetworkX backend API.
- Impact: resolved real frontend/backend integration issues (CORS); delivered an interactive, app-like simulator.

WORK EXPERIENCE

Data Modeler

Jul 2025 – Present

- Modeled business processes and data structures (ERD/MER), giving direct context on how backend data should be shaped for clean consumption in front-end interfaces.
- Documented workflows with swimlane diagrams and MRDs, reducing ambiguity between data, backend, and UI requirements.

Web Developer

Dec 2024 – Jun 2025

- Built and customized client-facing websites on Odoo and Shopify, editing front-end templates and components to match business and UX requirements.
- Gathered requirements directly from clients and iterated on UI/UX based on feedback, shipping changes in production.

- Handled deployments and incident resolution, minimizing client-facing downtime.

EDUCATION

Universidad Autónoma de Occidente — B.S. in Computer Science and Systems Engineering (2023 – present)

Servicio Nacional de Aprendizaje (SENA) — Technician in Multimedia Design and Integration (2022 – 2023)

LANGUAGES & STRENGTHS

Conversational English · Logical & systematic thinking · Assertive communication · Fast learner · Adaptability · Attention to detail · Proactivity & initiative · Cross-functional collaboration